

13/04/2025

Exercise: Given the following database

- **CITY**(ID, Name, #CountryCode, District, Population),
- **COUNTRY**(Code, Name, Continent, Region, SurfaceArea, IndepYear, Population, LifeExpectancy, GNP, GNPOld, LocalName, GovernmentForm, HeadOfState, Capital, Code2)
- **COUNTRYLANGUAGE**(Language , #CountryCode, , IsOfficial, Percentage)

Write the following SQL queries:

1. Find the list of official languages (IsOfficial = 'T').
2. Find the countries for which Arabic is an official language (IsOfficial = 'T').
3. Find the population density of all countries.
4. Find the list of countries where the attribute IndepYear is unknown.
5. Count the number of tuples (number of cities) in the **city** table.
6. Find the number of values for the attribute IndepYear
7. Find the number of distinct values for the attribute IndepYear.
8. Find the list of countries (code, name, indepyear, population) ordered by the indepyear attribute in descending order.
9. Find the list of countries (code, name, indepyear, population) ordered by indepyear descending and population ascending.
10. Find the capital name and president name of the countries *Algeria* and *Egypt*.
11. Find the sum of populations of Syrian cities (countrycode = 'SYR').
12. Find the minimum, maximum, and average values of the population attribute from the **country** table.
13. Find the total number of cities in each country and the sum of their populations (using grouping), and sort the result in descending order by the attribute countrycode.
14. For each country having more than 200 cities, find the number of cities and the sum of their populations.
15. Find all cities whose population is greater than the population of at least one Algerian city (code = 'DZD').
16. Find all cities whose population is greater than the population of all Chinese cities.
17. Find the languages spoken in *France* (name = 'France') (using subqueries).
18. Find the list of all languages whose percentage is higher than the average percentage, and indicate by how much they exceed the average percentage.
19. List the languages spoken in the African continent and located in a region that contains the letter 'N'.